FLEN: A Conflict-Aware Federated Framework for Cross-Jurisdiction Legal Language Modeling

Robert J.[a]

[a]*Independent Research Project,*

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ABSTRACT

Recent advances in legal large language models and federated adaptation support collaborative legal information processing while keeping institutional case records local. However, existing pipelines often interpret predictive disagreement as legal conflict without considering differences in applicable authority. They also provide limited control over local adaptation that can degrade model performance. Moreover, they often lack a coordinated process for verifying generated arguments before reaching a final judgement. To address these limitations, we propose the Federated Legal Expertise Network (FLEN). First, FLEN checks whether client outputs are comparable under the target authority and weights compatible contributions according to their conflict exposure. Second, it separates shared and private adaptation into independent LoRA branches and uses held-out local records to select private updates. Third, distinct agents develop and challenge legal arguments, verify the supporting evidence, and synthesize a final judgement. Experiments on CaseHOLD compare FLEN components with FedAvg, FedProx, FedAdam, and Ditto under matched protocols. Across five seeds, gating improves test accuracy over unguarded personalization by 2.50 and 2.64 percentage points at two fixed training budgets. MultiEURLEX experiments additionally include unadapted and local-only references. Proxy-FLEN achieves mean micro-F_1 of 0.336, compared with 0.337 for FedAvg and 0.310 for the frozen-backbone reference. These results show that gating improves personalization while proxy-FLEN maintains multilingual performance close to FedAvg.

1. Introduction

Legal information processing increasingly depends on models that connect facts, statutes, precedents, citations, and procedural context. Large language models (LLMs) are therefore attractive for legal judgement prediction, contract review, legal question answering, case retrieval, and statutory interpretation. Yet the dominant development pattern remains centralized: case documents, annotations, citations, and reasoning traces are collected into one training environment. This assumption is poorly matched to judicial practice because courts, law firms, public agencies, and legal research institutions may be unable to pool records containing personal information, confidential filings, sealed evidence, or jurisdiction-restricted data.

The technical challenge is that data locality and jurisdictional heterogeneity interact. Legal corpora are not merely statistically non-IID; they encode different sources of authority, procedural norms, citation conventions, languages, tasks, and decision structures. A model trained on pooled or indiscriminately averaged data can therefore learn majority-jurisdiction shortcuts or transfer a rule beyond its authority boundary. Conventional federated learning keeps raw data local while aggregating client updates [1, 2], but a single global objective does not distinguish task mismatch, statistical drift, model error, and legally legitimate disagreement. That distinction is the central technical problem studied here.

This paper studies cross-jurisdiction legal language modeling as a federated information-processing problem whose outputs must support audit of authorities, reasoning status, and disclosure paths. We propose the Federated Legal Expertise Network (FLEN), in which each institution performs parameter-efficient local adaptation and retains jurisdiction-specific knowledge, while the server transfers information judged compatible with a specified target task and authority context. A typed multi-agent workflow separates issue parsing, jurisdiction alignment, argument and counterargument generation, citation verification, conflict detection, policy auditing, and final synthesis.

The paper makes three contributions. First, it defines authority-scoped comparability and target-compatible aggregation, separating legal transfer eligibility from predictive disagreement. Second, it introduces independent shared and private LoRA branches with a local holdout gate, and isolates the gate's contribution through two matched training

budgets. Third, it connects exported federated adapters to a typed deliberation workflow and evaluates it against one-call and four-call single-agent controls. Reproducible CaseHOLD, MultiEURLEX, privacy, and published-label legal diagnostics characterize the benefits and costs of these components.

The experiments address four research questions. **RQ1** asks whether target-conditioned aggregation executes as designed and changes client contributions when proxy conflict exposure differs. **RQ2** asks which part of the shared/local training schedule determines aggregate and worst-client performance under non-IID CaseHOLD partitions. **RQ3** asks how source-language federation compares with unadapted and target-supervised references within a shared EU legal authority. **RQ4** asks whether client-update noise and typed deliberation improve their respective privacy and reasoning diagnostics without unacceptable utility loss. Section 5 reports the corresponding evidence.

1.1. Legal-theoretical motivation

Cross-jurisdiction legal reasoning differs from ordinary domain adaptation because disagreement can be normatively legitimate. Legal-pluralist scholarship shows that multiple legal orders may govern related social facts, while the operation and authority of those orders cannot be reduced to a single universal hierarchy [3, 4]. A statutory article, precedent, treaty provision, and contractual clause may concern related facts yet derive force from different institutions. Precedent hierarchy further determines whether a citation is binding, persuasive, or irrelevant within a target jurisdiction, and legally acceptable reasoning may therefore differ even when surface facts are similar.

These properties imply that legal heterogeneity is not merely a sampling artifact. Forcing all clients toward a single average can erase legitimate legal diversity, yet treating every difference as protected diversity can preserve model error or unsupported authority. The aggregation problem is therefore conditional: transfer shared representations only when task and authority scopes are comparable, preserve jurisdiction-specific components when they are not, and abstain when the available evidence cannot resolve the distinction. This requirement motivates the comparability gate, conflict diagnostics, and target-conditioned aggregation developed below.

2. Related Work

2.1. Legal language models and legal benchmarks

Legal natural language processing has moved from task-specific encoders toward domain-adapted transformer and LLM systems. Legal-BERT showed that legal-domain pretraining improves downstream legal understanding tasks [5], while long-document models such as Lawformer target the long-context structure of legal cases [6]. More recent legal LLMs and legal assistants, including SaulLM, Lawyer-LLaMA, and ChatLaw, adapt instruction-following or role-specialized models to legal tasks and legal corpora [7, 8, 9]. Benchmark suites such as LexGLUE, CaseHOLD, ECtHR, CUAD, CAIL, LeCaRD, and LawBench support evaluation across legal classification, case retrieval, contract review, and judgement prediction [10, 11, 12, 13, 14, 15, 16]. Recent work on cross-jurisdiction legal case summarization further shows that legal summarization systems face transfer barriers across legal systems, and rule-understanding benchmarks such as CLEAR highlight that LLMs must distinguish confusing legal provisions rather than only retrieve surface-similar statutes [17, 18]. These resources establish the empirical base for legal AI, but they are normally used in centralized settings. Our framework instead treats each corpus or institution as a potentially private jurisdictional client.

2.2. Federated learning under heterogeneous clients

Federated averaging introduced communication-efficient decentralized model training without sharing raw data [1]. FedProx, SCAFFOLD, and FedNova address heterogeneous optimization dynamics [19, 20, 21], while clustered and personalized methods retain multiple client optima rather than forcing one global model [22, 23]. Flower provides a flexible orchestration framework [24], and recent work studies federated LLM fine-tuning and heterogeneous low-rank adaptations [25, 26, 27]. These lines are the closest algorithmic comparators for FLEN: a legal-conflict method must outperform generic clustering or personalization on matched tasks and must report worst-jurisdiction behavior, not only a global average.

2.3. Parameter-efficient, private, and multi-agent legal reasoning

Full fine-tuning of legal LLMs is expensive and communication-heavy. LoRA reduces adaptation to low-rank matrices and makes federated exchange more feasible [28]. Differential privacy and secure aggregation provide distinct statistical and cryptographic protections [29, 30], but neither follows automatically from federated training. Legal

Table 1
Positioning relative to adjacent systems. A "Yes" indicates that the property is an explicit design target; "Partial" indicates indirect support without a dedicated mechanism.

System family	Federated	Legal AI	Multi-agent reasoning	Conflict-aware aggregation	Cross-jurisdiction design
FedAvg / FedProx / SCAFFOLD	Yes	No	No	No	No
Legal-BERT / Lawformer / SaulLM	No	Yes	No	No	Partial
Federated LLM fine-tuning / FLoRA	Yes	No	No	No	No
FEDLEGAL / FedJudge	Yes	Yes	No	No	Partial
Role-based legal agent workflow	No	Yes	Yes	No	Partial
FLEN (ours)	Yes	Yes	Yes	Yes	Yes

NLP work has studied differentially private language modeling, and recent cross-jurisdiction legal sentiment analysis combines federation with differential privacy [31, 32]. Conversely, communicated federated LLM updates can leak client-exclusive information [33]. These findings motivate an explicit threat model and attack evaluation; adapter exchange or a sanitization label alone is not a privacy guarantee. Multi-agent systems provide a separate inference-time abstraction. LangGraph coordinates stateful workflows [34], and legal applications benefit from separating argument generation, citation checking, contradiction detection, and final synthesis because LLMs can produce plausible but unsupported authorities [35].

2.4. Distinction from adjacent systems

Federated legal NLP already provides substantive baselines. FEDLEGAL offers naturally partitioned court-level benchmarks and shows that real non-IID legal data are harder than heuristic splits [36]; FedJudge combines parameter-efficient legal LLM adaptation with federated and continual-learning components [37]. FLEN contributes an authority-scoped treatment of conflict: disagreement is validated, routed to a compatible transfer cohort, retained locally, or escalated to abstention. Table 1 locates this mechanism relative to FEDLEGAL/FedJudge-style training, clustered and personalized FL, and role-based legal-agent workflows; the evaluation protocol specifies matched comparisons among them.

3. Motivation

3.1. Why federated learning is needed beyond multi-agent collaboration

Federated learning and multi-agent collaboration solve complementary problems. Modern LLM agents coordinate specialized roles through routing, debate, tool use, and task decomposition. In legal applications, institutional agents can exchange arguments, retrieved authorities, and intermediate reasoning summaries while raw case files remain local.

The two paradigms operate at different stages. Multi-agent systems organize inference-time reasoning: they divide a query, challenge analyses, verify citations, and synthesize an answer. Federated learning supports training-time collaboration under data-sharing constraints by improving shared representations or adapters from distributed private experience. Agent collaboration governs how legal expertise is used in a case; federated learning governs how that expertise is accumulated across cases and institutions.

Institutional expertise also reflects long-run exposure to local case distributions, procedural habits, evidentiary preferences, citation practices, and jurisdiction-specific interpretation patterns. These tacit statistical signals are learned across cases. Multi-agent workflows organize consultation among models at inference time; federated learning updates their competence from distributed institutional data.

FLEN combines these paradigms across training and inference. Federated learning supplies a data-local substrate for updating institutional adapters, while multi-agent deliberation selects, challenges, verifies, and synthesizes local outputs for a query. FLEN thereby specifies when institutional knowledge transfers, remains jurisdiction-specific, or triggers abstention. Formal privacy is supplied by the separate protocol in Section 4.7.

4. Methodology

4.1. Problem formulation

Let $\mathcal{K} = \{1, \dots, K\}$ denote participating legal institutions. Client k owns a private legal dataset $\mathcal{D}_k = \{(x_i, y_i, c_i, j_i)\}_{i=1}^{n_k}$, where x_i is a case or legal query, y_i is the task label or target answer, c_i denotes citation metadata,

and j_i denotes jurisdictional metadata. Raw samples in $\mathcal{D}_k$ never leave client k. A global legal model with parameters w is adapted through local parameter-efficient updates Δ_k, such as LoRA adapters. The standard federated objective is

$$\min_{w} \sum_{k=1}^{K} p_k \mathcal{L}_k(w), \quad p_k = \frac{n_k}{\sum_{\ell=1}^{K} n_\ell}. \tag{1}$$

When supervision is available, local adaptation may include task-specific legal consistency terms:

$$\mathcal{L}_k^{\text{client}} = \mathcal{L}_k^{\text{task}} + \beta_{\text{cit}} \mathcal{L}_k^{\text{citation}} + \beta_{\text{align}} \mathcal{L}_k^{\text{alignment}} + \beta_{\text{debate}} \mathcal{L}_k^{\text{debate}}. \tag{2}$$

Here, $\mathcal{L}_k^{\text{citation}}$ uses gold or independently verified authorities, $\mathcal{L}_k^{\text{alignment}}$ uses an audited cross-jurisdiction rule mapping, and $\mathcal{L}_k^{\text{debate}}$ uses labelled argument–counterargument data. Each coefficient is active precisely when its supervision is available, keeping the trained objective aligned with observed evidence.

We further define a cross-client legal conflict score for comparable query q and client pair (i, j):

$$D_{\text{conflict}}(i, j; q) = \omega_{\text{cit}} D_{\text{cit}}(i, j; q) + \omega_{\text{rea}} D_{\text{rea}}(i, j; q) + \omega_{\text{ver}} D_{\text{ver}}(i, j; q) + \omega_{\text{rule}} D_{\text{rule}}(i, j; q), \tag{3}$$

where D_{cit} measures citation divergence, D_{rea} measures reasoning-chain contradiction, D_{ver} measures incompatible verdict distributions under comparable facts, and D_{rule} measures rule-alignment distance. The coefficients satisfy $\omega_x \geq 0$ and $\sum_x \omega_x = 1$.

4.2. Comparability gate and probe construction

Conflict is meaningful only when two client outputs address the same legal issue in a compatible output and authority space. At round t, the server therefore uses a probe set $\mathcal{P}_t$ that is disjoint from all client training, validation, and test records. A probe may be a licensed public case, an expert-authored hypothetical, or a fact pattern sanitized before federation; its provenance, target task, target jurisdiction, language, label ontology, and admissible authority scope must be recorded. Evaluation examples are never reused as aggregation probes.

For clients i and j and probe q, let $a_{ij}(q) \in \{0, 1\}$ be a comparability indicator. It equals one when both clients address a compatible task and their outputs can be interpreted under the same target jurisdiction or a validated authority mapping. Differences with $a_{ij}(q) = 0$ are not penalized; they remain jurisdiction-specific. This gate separates potentially harmful inconsistency from task mismatch or legally legitimate scope differences.

The full authority-routing protocol assigns each comparable pair one of four states using a classifier calibrated on expert labels: TRANSFERABLE evidence may enter the shared cohort; AUTHORITYLOCAL evidence is valid only in its original authority scope and remains local; UNSUPPORTED evidence may be downweighted or rejected; and UNRESOLVED evidence triggers conservative weighting or abstention. Authority eligibility is frozen from audited metadata before training, whereas model-dependent unsupported and unresolved probabilities may be updated on disjoint probes. The completed training studies instantiate routing with automatic metadata and gold-error-gated verdict disagreement. Expert calibration of these four legal categories remains a separate construct-validation study.

4.3. Operational legal conflict functions

For a comparable probe q evaluated locally by clients i and j, client k returns a minimized tuple

$$s_k(q) = \left(C_k(q),\ \rho_k(q),\ \boldsymbol{r}_k(q),\ \boldsymbol{\pi}_k(q),\ \mathcal{R}_k(q)\right), \tag{4}$$

where $C_k(q)$ is a canonicalized authority set, $\rho_k(q)$ is a disclosed reasoning summary, $\boldsymbol{r}_k(q)$ is its embedding, $\boldsymbol{\pi}_k(q)$ is the predicted verdict distribution, and $\mathcal{R}_k(q)$ is a typed set of applicable rule identifiers with authority levels. This minimized tuple remains within the leakage threat model specified in Section 4.7.

Citation conflict. Let C_i and C_j be normalized citation sets after statute/article/case canonicalization into the target authority space. For a comparable pair with at least one citation, citation conflict is the Jaccard distance:

$$D_{\text{cit}}(i, j; q) = 1 - \frac{|C_i(q) \cap C_j(q)|}{|C_i(q) \cup C_j(q)|}. \tag{5}$$

If both sets are empty, citation conflict is marked unavailable rather than zero, so omission is not rewarded. If the cited authorities belong to different valid jurisdictional scopes, $a_{ij}(q) = 0$ unless an audited mapping projects them into a common target authority space. A soft graph-overlap score may be reported as a secondary diagnostic, but it must be validated separately from exact citation identity.

Reasoning conflict. Each minimized reasoning summary is encoded as $\boldsymbol{r}_k(q) \in \mathbb{R}^d$. A legal natural-language-inference classifier g_{NLI} produces a directional contradiction probability from paired summaries. We symmetrize that score:

$$D_{\mathrm{rea}}(i,j;q) = \eta \frac{1-\cos(\boldsymbol{r}_i(q),\boldsymbol{r}_j(q))}{2} + \frac{1-\eta}{2}\left[g_{\mathrm{NLI}}(\rho_i(q),\rho_j(q)) + g_{\mathrm{NLI}}(\rho_j(q),\rho_i(q))\right], \tag{6}$$

where $\eta \in [0,1]$ trades geometric dissimilarity against explicit contradiction evidence. If no validated legal NLI model is available, the embedding term must be reported alone and the limitation disclosed.

Verdict conflict. For verdict distributions $\boldsymbol{\pi}_i(q)$ and $\boldsymbol{\pi}_j(q)$ over the same label space or an audited common ontology, we use Jensen–Shannon divergence with base-2 logarithms, which bounds the score in $[0,1]$:

$$D_{\mathrm{ver}}(i,j;q) = \mathrm{JSD}(\boldsymbol{\pi}_i(q),\boldsymbol{\pi}_j(q)) = \frac{1}{2}\mathrm{KL}(\boldsymbol{\pi}_i\|\boldsymbol{m}) + \frac{1}{2}\mathrm{KL}(\boldsymbol{\pi}_j\|\boldsymbol{m}), \quad \boldsymbol{m} = \frac{\boldsymbol{\pi}_i+\boldsymbol{\pi}_j}{2}. \tag{7}$$

Before evaluation, zero entries are replaced by a fixed ϵ and renormalized. Unlike label disagreement, this score retains uncertainty in close or indeterminate cases.

Rule-alignment distance. We represent applicable legal authorities as a typed rule graph $\mathcal{G}_k = (\mathcal{R}_k, E_k)$ with nodes for statutes, precedents, clauses, or treaty provisions and typed edges for hierarchy, exception, citation, or implementation relations. Let $M_{ij}(q)$ be a one-to-one node matching audited for the target jurisdiction. Node mismatch is

$$D_{\mathrm{node}}(i,j;q) = 1 - \frac{2}{|\mathcal{R}_i(q)| + |\mathcal{R}_j(q)|} \sum_{(u,v)\in M_{ij}(q)} \mathrm{sim}_{\mathrm{rule}}(u,v), \tag{8}$$

where $\mathrm{sim}_{\mathrm{rule}}(u,v) = \mathbb{I}_{\mathrm{auth}}(u,v)\max\{0,\cos(\boldsymbol{e}_u,\boldsymbol{e}_v)\}$ combines semantic similarity with authority compatibility. Let D_{edge} be the fraction of mapped typed edges whose relation or direction is not preserved. The complete rule distance is $D_{\mathrm{rule}} = (1-\zeta)D_{\mathrm{node}} + \zeta D_{\mathrm{edge}}$, with ζ calibrated on expert annotations. Probes without an admissible rule mapping are excluded from this component and reported separately.

Let $\mathcal{E}_t = \{(i,j,q) : q \in \mathcal{P}_t,\ i<j,\ a_{ij}(q)=1\}$ be the valid comparison set after removing unavailable components. The per-round server score is

$$\overline{D}_x^{(t)} = \frac{1}{|\mathcal{E}_{t,x}|} \sum_{(i,j,q)\in\mathcal{E}_{t,x}} D_x(i,j;q), \quad x \in \{\mathrm{cit},\mathrm{rea},\mathrm{ver},\mathrm{rule}\}. \tag{9}$$

Each component reports its own denominator $|\mathcal{E}_{t,x}|$ and confidence interval. The four weights ω_x and all decision thresholds are calibrated on a held-out, expert-labelled conflict set rather than on the task test data. When a conflict component is unavailable, its weight is removed and the remaining available component weights are renormalized for that pair; unavailable evidence is never imputed as agreement. Calibration reports class-conditional AUROC/AUPRC, expected calibration error, coverage, and abstention frequency. The probe-development, calibration, and final evaluation sets are immutable and mutually disjoint so that conflict thresholds cannot be tuned on reported task outcomes.

4.4. Federated legal LLM architecture

The system follows five layers: a federated judicial data layer, a local legal LLM training layer, a jurisdiction-aware representation layer, a conflict-aware aggregation layer, and a federated multi-agent reasoning layer. Figure 1 organizes these layers into a training-time federated loop and an inference-time verified reasoning path.

Each institution maintains local case corpora, citation graphs, legal knowledge bases, and reasoning trajectories. The design distinguishes jurisdiction, legal domain, language, institution type, and court level rather than treating

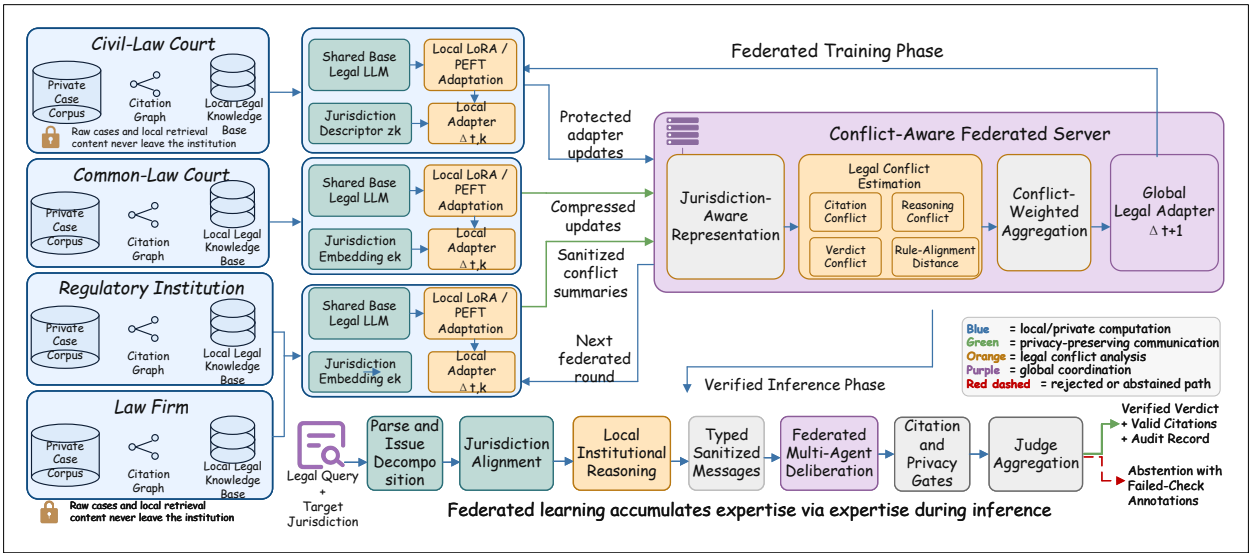


Figure 1: FLEN: Federated Legal LLM Architecture. Institutions retain raw records and local retrieval state while exchanging configured adapter payloads and minimized probe summaries. The method applies comparability and authority checks before target-conditioned aggregation, retains guarded jurisdiction-local residuals, and gates inference outputs by citation, conflict, and policy status. The CaseHOLD screening exercises the training-time path with automatic proxy metadata; Sections 4.7 and 5 define the privacy boundary and report the measured studies.

them as interchangeable client labels. Non-IID simulation may vary topic or label concentration within a task; a cross-jurisdiction experiment additionally requires authentic jurisdiction metadata and a common task or audited output mapping.

Each client adapts Qwen2.5-0.5B-Instruct using LoRA. The measured configuration uses rank 8, scaling factor 16, dropout 0.05, and target modules `q_proj` and `v_proj`. Payload type is fixed within each controlled comparison. Gradients, adapters, and reasoning embeddings remain distinct communication objects in the system design.

The two paths in Figure 1 clarify the complementary roles of federation and agent collaboration. The upper path accumulates jurisdiction-sensitive capability from distributed private experience and broadcasts the global adapter after conflict-aware weighting. The lower path applies that capability to a particular query: institutional outputs are converted into admissible typed messages, challenged through deliberation, and released only after citation and privacy verification. Consequently, the multi-agent layer does not replace federated learning, and the federated server does not directly synthesize an unverified judicial decision.

4.5. Jurisdiction-aware representation

Legal non-IID data cannot be represented only by label distributions. We attach a structured descriptor z_k to each client containing jurisdiction, legal domain, language, institution type, court level, and authority hierarchy. In the full method, categorical fields are embedded and concatenated with a text encoding of the documented jurisdiction profile to form $e_k \in \mathbb{R}^d$. The descriptor conditions local training and target-jurisdiction routing:

$$h_i = f_\theta(x_i, e_k, r_i), \tag{10}$$

where r_i denotes retrieved local legal context and h_i is the case representation used for prediction or reasoning. The descriptor schema, encoder version, training objective, and missing-value policy are frozen before evaluation. The current pilot uses recorded metadata for routing; jurisdiction-embedding evaluation belongs to the confirmatory protocol.

4.6. Conflict-aware aggregation

Standard FedAvg updates a single global model by weighted averaging:

$$w_{t+1}^{\text{FedAvg}} = \sum_{k=1}^{K} p_k w_{t,k}, \quad p_k = \frac{n_k}{\sum_{\ell=1}^{K} n_\ell}. \tag{11}$$

Unconditional conflict downweighting would suppress a legally valid minority client whenever it disagrees with a larger aligned group. FLEN therefore conditions transfer on a target jurisdiction j and retains a jurisdiction-local residual. Let $\mathcal{E}_{t,k}^{(j)}$ contain valid comparisons involving client k that passed the comparability gate for target j. The unresolved conflict exposure is

$$\delta_{k\to j}^{(t)} = \frac{1}{|\mathcal{E}_{t,k}^{(j)}|} \sum_{(k,\ell,q)\in\mathcal{E}_{t,k}^{(j)}} D_{\text{conflict}}(k,\ell;q), \tag{12}$$

with $\delta_{k\to j}^{(t)} = 0$ when no comparison is admissible. Let $u_{k\to j} \in [0,1]$ be the frozen target-authority compatibility score and let $n_{\max}$ cap the influence of client size. The normalized base weight is

$$b_{k\to j} = \frac{u_{k\to j}\min(n_k, n_{\max})}{\sum_{\ell\in\mathcal{G}_j} u_{\ell\to j}\min(n_\ell, n_{\max})}. \tag{13}$$

Citation volume or agreement with a majority is not used as a reliability multiplier unless its calibration is independently validated. A conflict-sensitive candidate weight is

$$\widehat{\alpha}_{k\to j}^{(t)} = \frac{b_{k\to j}\exp(-\lambda\delta_{k\to j}^{(t)})}{\sum_{\ell\in\mathcal{G}_j} b_{\ell\to j}\exp(-\lambda\delta_{\ell\to j}^{(t)})}, \tag{14}$$

where $\mathcal{G}_j$ is the transfer-compatible cohort. To prevent a feedback loop that eliminates a minority jurisdiction, FLEN mixes this candidate with the base distribution using a diversity floor $\tau \in [0,1]$:

$$\widetilde{\alpha}_{k\to j}^{(t)} = (1-\tau)\widehat{\alpha}_{k\to j}^{(t)} + \tau b_{k\to j}. \tag{15}$$

For stability, deployed weights use a pre-specified exponential update

$$\alpha_{k\to j}^{(t)} = (1-\nu)\alpha_{k\to j}^{(t-1)} + \nu\widetilde{\alpha}_{k\to j}^{(t)}, \tag{16}$$

followed by normalization within $\mathcal{G}_j$. The completed CaseHOLD revision fixes $\nu = 0.5$, $\lambda = 0.2$, and $\tau = 0.1$ before final testing. The zero-conflict control uses $\lambda = 0$. These settings evaluate the specified mechanism; validation-jurisdiction selection and a coefficient-sensitivity study belong to the full authority-routing protocol. The implementation aggregates the shared LoRA factor state $\vartheta_s = (A_s, B_s)$ within the target cohort. Its shared state and effective client-layer weight are

$$\vartheta_{s,t+1,j} = \sum_{k\in\mathcal{G}_j} \alpha_{k\to j}^{(t)}\vartheta_{s,t,k}, \qquad W_{t+1,k} = W_0 + s\big(B_s A_s + \gamma_k B_k A_k\big), \tag{17}$$

where s is the LoRA scale. Factor averaging is the shared aggregation rule for the compared PEFT methods; each resulting factor state defines its effective matrix update. The revised implementation uses two independent LoRA branches with identical target modules. For a layer input h, its adapter output is $B_s A_s h + \gamma_k B_k A_k h$, with the usual LoRA scaling and a retained-branch indicator $\gamma_k \in \{0,1\}$. The shared branch is initialized from the target-cohort adapter and trained during the federated phase; its factors alone are uploaded. The private branch persists client-side and is optimized with the shared factors frozen. A deterministic holdout reserved from each client's training shard is excluded from both training phases for every compared method. The private branch is retained when the composed model preserves holdout accuracy and loss; otherwise inference uses the shared branch alone and the private output is reset to zero. External validation and test sets never participate in this selection. Adding two low-rank matrix outputs

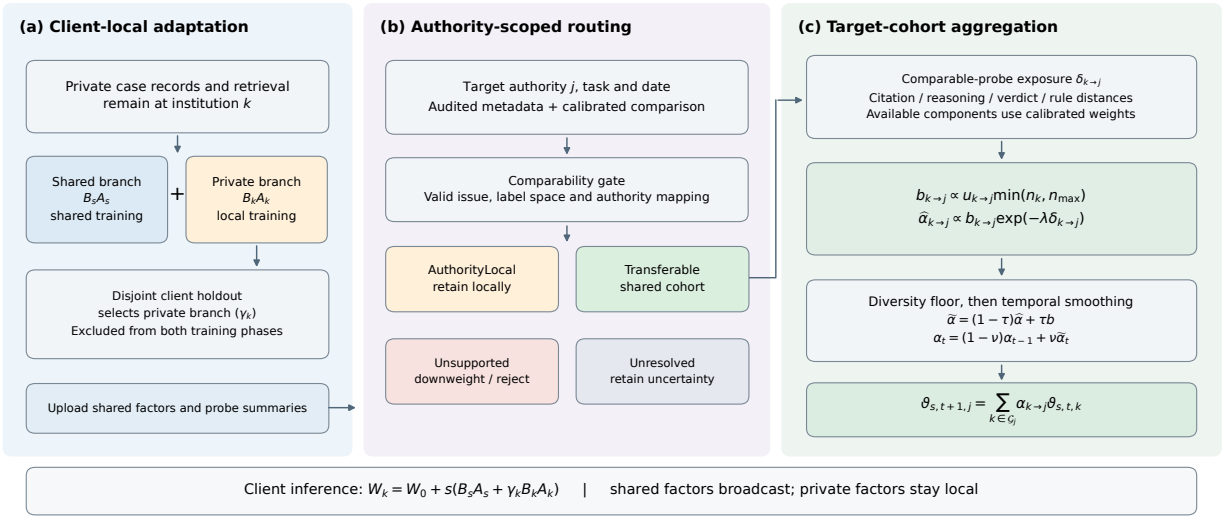


Figure 2: Target-conditioned legal-conflict routing and independent LoRA branches. (a) Shared and private branches have separate factors; the private branch is selected on a disjoint client holdout. (b) Audited authority scope and comparability determine transfer routing. (c) Comparable-probe exposure sets cohort weights, followed by a diversity floor and temporal smoothing. Shared factors are aggregated and broadcast; private factors remain local. The full protocol uses expert-calibrated authority routing, while the screening evaluates automatic proxy exposure.

Algorithm 1: Conflict-aware federated legal LoRA aggregation

Input: Clients $\mathcal{K}$, target cohorts, probe set, rounds T, conflict weight λ, diversity floor τ
Output: Target-conditioned shared adapters and jurisdiction-local residuals

for $t = 0$ **to** $T-1$ **do**
  Server broadcasts target-cohort adapters, probe identifiers, and the frozen configuration
  foreach *client k in parallel* **do**
    Client trains shared factors $\vartheta_{s,t,k}$ on $\mathcal{D}_k \setminus H_k$ with the private branch disabled
    Client computes available minimized conflict summaries from the shared branch on disjoint probes
    Client trains its private branch on $\mathcal{D}_k \setminus H_k$ with shared factors frozen
    Client selects the private branch on H_k and keeps its factors local
    Client sends shared factors and configured probe summaries under the declared threat model
  end
  Server computes $D_{\text{cit}}, D_{\text{rea}}, D_{\text{ver}}, D_{\text{rule}}$
  Server applies the comparability gate and computes target-conditioned exposure $\delta_{k\to j}^{(t)}$
  Server forms diversity-floored, temporally smoothed weights and aggregates shared factors per target cohort
  Server logs communication cost, client drift, aggregation stability, and privacy metrics
end

differs from adding their factors: $(B_s + B_k)(A_s + A_k)$ introduces cross terms. The legacy screening used factor-space residuals; the revised experiments evaluate the independent-branch implementation. Figure 2 illustrates the aggregation mechanism.

Figure 2 summarizes the training-time mechanism: transfer eligibility determines the cohort, and comparable-probe exposure determines its shared weights. The completed studies execute this path with automatic verdict proxies and independent LoRA branches. Citation, reasoning, and rule-alignment scores define the expert-calibrated extension.

4.7. Threat model and privacy boundary

FLEN's implemented baseline provides data locality: raw client records are not intentionally uploaded to the server. Adapters, gradients, verdict distributions, embeddings, summaries, and audit logs remain potential channels for membership, attribute, or text leakage. The policy auditor and `privacy_tag` record requested controls and schema-check outcomes; formal protection is defined by the differential-privacy and cryptographic protocol below.

A privacy-preserving instantiation specifies the adversary, protected unit, adjacency relation, clipping norm, sampling rate, noise multiplier, accountant, and final (ϵ, δ) after composition. The completed experiment applies a central Gaussian mechanism to clipped client updates under a trusted aggregator and evaluates a black-box record-membership attack. Its protected unit is a client update; its attack target is an individual source record. Extending protection to disclosed probe summaries requires a separate privacy budget or cryptographic computation. Update reconstruction and inference-message recovery are additional attack studies in the full protocol.

Secure aggregation also changes the conflict protocol. A compatible two-stage design first computes auditable weights from public probes and differentially private summaries, or through secure multiparty computation; clients then scale their updates by the published weights and securely aggregate the weighted sum. This proposed cryptographic extension requires dropout, collusion, and overhead evaluation. The measured Gaussian mechanism protects aggregate releases under its stated trusted-aggregator assumptions.

4.8. Federated multi-agent legal reasoning protocol

The proposed multi-agent workflow is represented as a directed state graph $\mathcal{G}_A = (V_A, E_A)$. Its state nodes are

$$V_A = \{\textsc{Parse}, \textsc{Align}, \textsc{Local}, \textsc{Argue}, \textsc{Counter}, \textsc{Verify}, \textsc{Detect}, \textsc{Audit}, \textsc{Judge}\}.$$

A query moves through typed state transitions rather than unrestricted prompt exchange. A transmitted message has schema

$$m = (\text{case_id}, \text{sender}, \text{jurisdiction}, \text{issue_ids}, \boldsymbol{C}, \rho, \boldsymbol{r}, \boldsymbol{\pi}, \text{privacy_tag}), \tag{18}$$

where $\boldsymbol{C}$, ρ, $\boldsymbol{r}$, and $\boldsymbol{\pi}$ are the minimized summaries used by the conflict functions. The `privacy_tag` records requested controls and realized budget identifiers. Messages containing raw case text or client-local retrieval content are not admissible on inter-client edges, but the admissible fields remain subject to the leakage analysis in Section 4.7.

The protocol defines three memory scopes. Short-term deliberation memory $\boldsymbol{M}^{\text{case}}$ stores the active issue decomposition, disclosed argument summaries, contradiction flags, and the judge decision for a single query. Jurisdiction memory $\boldsymbol{M}_k^{\text{jur}}$ stores locally maintained rule mappings, authority hierarchies, and alignment metadata for client k; only sanitized mappings can be shared. Citation memory $\boldsymbol{M}^{\text{cit}}$ stores canonical citation identifiers, validity results, and authority-level checks required for reproducible citation consistency evaluation. A reasoning trace log records admissible message transitions and verifier decisions for later audit.

The argument and counterargument roles are assigned opposed issue-level analyses conditioned on the aligned jurisdiction state; task-specific labels may specialize them as prosecution/defence, claimant/respondent, or clause/support review. Before synthesis, the citation-verification stage rejects unsupported citations and incompatible authority claims. Conflict detection evaluates pairwise disclosed summaries and records unresolved conflicts. The judge stage receives verified arguments, verdict distributions, conflict scores, and policy-audit status, then emits a final output only when citation, authority, conflict, and privacy-policy gates are satisfied.

The measured inference implementation uses four sequential model calls for primary analysis, adversarial challenge, evidence review, and final judgement. These roles share one base or exported shared-adapter checkpoint and the same supplied CaseHOLD material. Evidence review is a prompted comparison with that material; external citation retrieval and authoritative source validation are components of the full graph. The automatic experiment measures answer accuracy, format compliance, and resource consumption; independent legal review evaluates soundness and unsupported claims.

Figure 3 makes the intended disclosure boundary explicit: private retrieval and jurisdiction memory remain client-local, whereas only schema-constrained messages enter shared deliberation. Case memory coordinates the active arguments, citation memory supports authority validation, and the reasoning trace records admissible transitions and gate outcomes. This separation is designed to let the judge synthesize institution-specific legal expertise without receiving unrestricted raw case content.

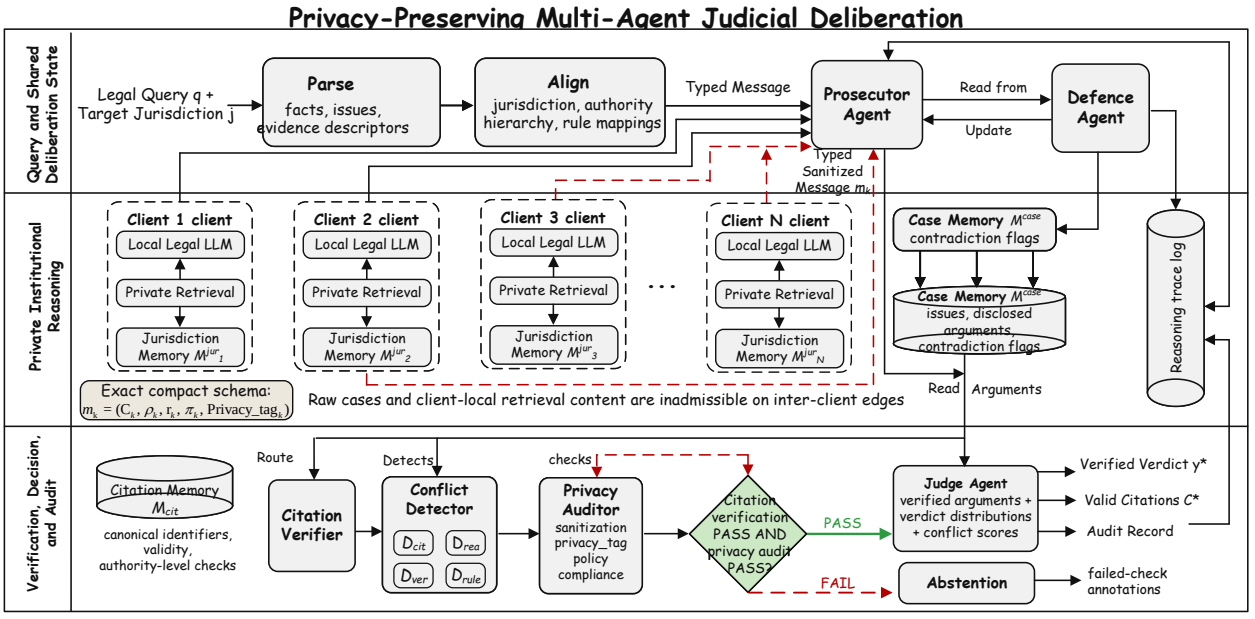


Figure 3: Proposed policy-gated data-local multi-agent deliberation. The workflow separates shared query state, client-local retrieval, and verification/decision/audit layers. A query is aligned with the target jurisdiction before institutions produce typed minimized messages. Argument and counterargument roles expose competing analyses, while citation, authority, conflict, and policy gates constrain admissible evidence. Formal privacy follows the protocol in Section 4.7.

Algorithm 2: Policy-gated data-local multi-agent judicial deliberation

Input: Query q, target jurisdiction j, local memories $\{M_k^{\text{jur}}\}$
Output: Verified verdict y^* with citations C^* and audit record

PARSE extracts issues and sanitized fact descriptors into M^{case}
ALIGN retrieves rule mappings and authority levels from M_j^{jur}
foreach *participating client* k **do**
  LOCAL emits admissible message $m_k = (C_k, \rho_k, \boldsymbol{r}_k, \boldsymbol{\pi}_k, \text{privacy_tag}_k)$
end
ARGUE and COUNTER construct opposed issue-level summaries
VERIFY canonicalizes citations and rejects invalid authority claims
DETECT computes $D_{\text{cit}}, D_{\text{rea}}, D_{\text{ver}}, D_{\text{rule}}$
AUDIT verifies sanitization and privacy-policy compliance
if *citation, authority, conflict-threshold, and privacy-policy checks pass* **then**
  JUDGE aggregates verified summaries and outputs (y^*, C^*)
end
else
  Return an abstention with failed-check annotations
end

5. Experiments

5.1. Experimental setup

The evaluation separates optimization, language transfer, privacy, legal components, and reasoning. CaseHOLD tests aggregation and personalization under controlled non-IID partitions. MultiEURLEX tests source-to-target language behavior within a shared EU regulatory authority. A client-update privacy study measures membership leakage and utility under replace-one Gaussian protection. LegalBench evaluates published-label legal components.

Table 2
Frozen revision protocols. Counts denote scheduled runs or conditions; completed outcomes are identified in the results.

Study	Matched design
CaseHOLD	Eight conditions × five seeds; twenty rounds; 1,200 training, 80 validation and 1,000 fixed test records; independent shared/private LoRA branches.
FedAdam	Five additional seeds; same ten-step clients as FedAvg; Flower adaptive server optimizer.
MultiEURLEX	Six methods × four holdouts × five seeds; three rounds; 500 training records and 64 separate probes per source client; 500 validation and 500 fixed test records per language.
Privacy	Five conditions × three seeds; unclipped, clipping-only and three noise levels; disjoint source probes and threshold validation.
LegalBench	Four tasks; 492 fixed cases; two pretrained checkpoints plus five FedAvg and five FLEN shared adapters.
Reasoning	Three checkpoints × two single-agent call budgets; same sixty cases; four-stage deliberation and blinded presentation.

Paired CaseHOLD inference compares single-agent prediction with typed four-stage deliberation. The federated studies use Qwen2.5-0.5B-Instruct with matched seeds or paired examples. Run-level means are accompanied by standard deviations; paired differences use 95% intervals over matched seeds or examples. The completed revision comprises 45 CaseHOLD, 120 MultiEURLEX, 15 privacy runs, and six paired reasoning packages. Eleven of twelve LegalBench checkpoint conditions have completed; the larger-model reference is pending.

The CaseHOLD comparison includes FedAvg, FedProx, Ditto, and FedAdam [38], independent shared/private branches with and without gating at two budgets, and proxy-FLEN with gating. MultiEURLEX compares FedAvg, FedProx, the $\lambda = 0$ control, proxy-FLEN, a frozen backbone with a supervised classification head, and a target-supervised local-only reference across four language conditions. The held-out language is excluded from federated threshold selection. Earlier factor-residual screenings remain available in the repository; the results below use the independent-branch revision.

All federated adapters use rank eight, scaling sixteen, and query/value projections. CaseHOLD uses AdamW at 2×10^{-4}, batch size one, accumulation over four batches, and a 512-token limit. MultiEURLEX uses AdamW at 2×10^{-5}, batch size four, a 256-token limit, and one local epoch per round, jointly training the classification head. CaseHOLD selects the holding with the smallest mean answer-token loss; MultiEURLEX applies a validation-selected threshold to 21 sigmoid label outputs.

Frozen tests and matched contrasts. The revised CaseHOLD study fixes 1,000 test records before training and evaluates them after round twenty. Four equal, fixed test shards define deployment clients. Guarded and unguarded personalization are compared at both 8+2 and 10+2 shared/private optimizer steps, with identical gate records excluded from both training phases. FedAdam uses the same ten-step clients as FedAvg and the Flower server optimizer with learning rate 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\tau = 10^{-9}$. Five matched training seeds determine run-level uncertainty. Paired Student-t intervals describe effect magnitude; exact two-sided sign-flip tests and Holm adjustment are reported for planned contrasts within each metric. With five pairs, the smallest attainable two-sided sign-flip p-value is 0.0625.

The revised MultiEURLEX matrix uses the same 500 test records per language across methods and training seeds. Each source client reserves 64 additional training-split records as aggregation probes, separate from the 500-record validation sample used for threshold selection. Test predictions are produced after the final round. For method m and seed s, average and worst held-out-language micro-F_1 are computed as

$$\bar{F}_{m,s} = \frac{1}{4}\sum_{\ell=1}^{4} F_{m,s,\ell}, \qquad F^{\min}_{m,s} = \min_{\ell \in \{1,\ldots,4\}} F_{m,s,\ell}. \tag{19}$$

Uncertainty and paired method differences are then calculated over training seeds. Thresholds at the search boundary remain in the analysis. Local-only adaptation uses target-language labels and is reported as a target-supervised reference.

Published-label component diagnostics. A separate LegalBench [39] protocol tests precedent overruling, UCC versus common-law applicability, citation support, and personal jurisdiction. A fixed sample contains 240, 94, 108,

Table 3
Independent dual-LoRA revision: five seeds, twenty rounds and 1,000 fixed test records. Accuracy entries are percent mean and seed SD; time is observed mean wall minutes.

Method (shared+private)	Accuracy	Worst client	Fit payload (MiB)	Minutes
FedAvg (10)	28.28 ± 0.77	26.80 ± 0.80	330.00	15.3
FedProx (10)	28.34 ± 1.09	26.40 ± 1.26	330.00	14.9
Ditto (8+2)	25.60 ± 0.70	22.48 ± 1.18	330.00	17.4
Unguarded (8+2)	25.28 ± 1.13	21.20 ± 2.06	330.01	40.1
Guarded (8+2)	27.78 ± 1.42	26.00 ± 1.52	330.01	38.0
Unguarded (10+2)	25.92 ± 1.55	22.32 ± 1.73	330.01	40.1
Guarded (10+2)	28.56 ± 1.03	26.40 ± 0.98	330.01	39.6
Proxy-FLEN (8+2)	27.84 ± 1.40	25.76 ± 1.56	330.01	38.7
FedAdam (10)	21.28 ± 0.61	18.96 ± 1.15	330.00	13.2

and 50 test cases, respectively. Each prompt supplies the legal material and allowed labels; citation classification also supplies the proposed citation. Labels are scored by mean answer-token negative log likelihood. The protocol compares the pretrained Qwen backbone with five FedAvg and five FLEN shared adapters, and includes pretrained Mistral-7B-Instruct-v0.2 as a larger-model reference. Accuracy, macro-F_1, balanced accuracy, multiclass Brier score, and ten-bin confidence ECE describe distinct outcomes. Adapter effects are paired across training seeds; the pretrained-reference contrast is paired across cases. These tasks measure benchmark components, while expert authority-pair annotation evaluates the legal-conflict construct.

Inference and resource controls. The revised reasoning protocol crosses base, FedAvg-shared, and FLEN-shared checkpoints with one-call and four-call single-agent controls and four-stage deliberation on the same sixty cases. The four-call control repeatedly rechecks its answer against the supplied holdings. Analysis prompt tokens, generated tokens, and elapsed time are recorded separately from the shared presentation step. A repeated deterministic four-stage output is counted once per checkpoint. Case-level bootstrap intervals and exact McNemar tests quantify prediction differences; the seed-41 checkpoint comparison measures those checkpoints. Logical federated payload and client drift accompany task performance. Elapsed times are interpreted alongside recorded GPU metadata.

5.2. Matched-budget dual-branch revision

The completed revision evaluates forty original configurations and five additional FedAdam runs. All runs use independent final-test predictions and the frozen training schedule. Worst-client accuracy is reconstructed from saved predictions using the same four round-robin test shards for every run; early shared-only runs originally logged a different grouping. Table 3 reports aggregate and worst-client outcomes.

At the $8 + 2$ budget, gating changes accuracy by +2.50 points (95% paired interval $[1.63, 3.37]$). At $10 + 2$, its effect is +2.64 points (95% paired interval $[1.60, 3.68]$). Adding proxy conflict weights to guarded $8 + 2$ changes accuracy by +0.06 points (95% paired interval $[-0.42, 0.54]$). These fixed-budget contrasts separate private selection from additional shared optimization.

The two gate effects are positive across all five seed pairs. Their exact two-sided sign-flip p-values are 0.0625, with Holm-adjusted $p = 0.4375$ across seven planned accuracy contrasts. The intervals quantify recovery relative to unguarded adaptation; the small seed count limits exact-test resolution. Guarded $8 + 2$ accuracy is 27.78%, compared with 28.28% for FedAvg and 28.34% for FedProx. FedAdam reaches 21.28% at the fixed server setting; its hyperparameters were not optimized on these test outcomes.

The gate retains 39.75% and 35.50% of client-round candidates at the two budgets. The selected private branch changes final accuracy relative to its shared branch by 0.00 and 0.16 points, respectively; unguarded private branches reduce it by 2.30 and 2.74 points. Thus the main observed benefit is removal of harmful private updates, while the retained private branch contributes little additional test utility.

Table 4
Private-branch diagnostics. Acceptance is the fraction of client-round candidates retained; the final test gain compares the selected composition with the shared branch, in percentage points.

Method	Accepted (percent)	Private minus shared
Unguarded (8+2)	100.00 ± 0.00	-2.30 ± 1.05
Guarded (8+2)	39.75 ± 10.44	0.00 ± 0.00
Unguarded (10+2)	100.00 ± 0.00	-2.74 ± 0.68
Guarded (10+2)	35.50 ± 5.97	0.16 ± 0.17
Proxy-FLEN (8+2)	38.25 ± 11.44	0.04 ± 0.09

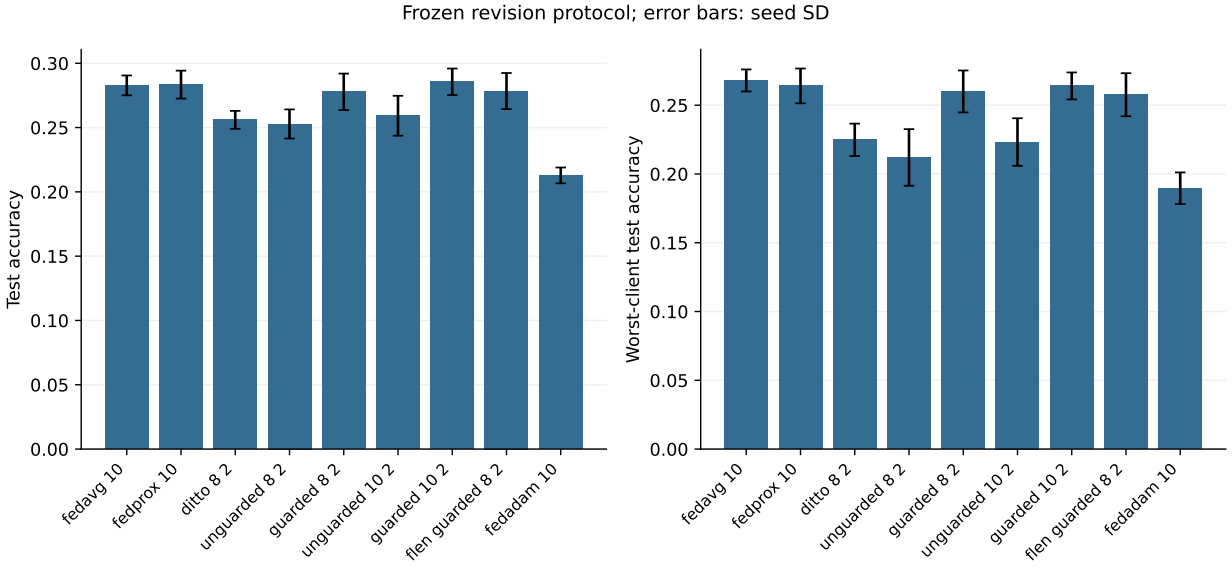


Figure 4: Completed dual-branch revision. Bars show five-seed means and seed SD on fixed test records. Worst client denotes a fixed test shard.

Table 5
Completed 120-run language-transfer revision. Each seed contributes a mean and minimum across four independently held-out target languages; entries report mean and seed SD.

Method	Mean micro F1	Worst micro F1	Macro F1	Brier
FedAvg	0.337 ± 0.015	0.273 ± 0.037	0.168 ± 0.010	0.184 ± 0.006
FedProx	0.335 ± 0.014	0.272 ± 0.037	0.167 ± 0.010	0.185 ± 0.006
$\lambda = 0$	0.337 ± 0.015	0.273 ± 0.037	0.168 ± 0.010	0.184 ± 0.006
Proxy-FLEN	0.336 ± 0.017	0.276 ± 0.036	0.169 ± 0.009	0.185 ± 0.006
Local only (target labels)	0.468 ± 0.011	0.449 ± 0.013	0.163 ± 0.011	0.121 ± 0.001
Frozen backbone + head	0.310 ± 0.013	0.256 ± 0.037	0.163 ± 0.009	0.214 ± 0.016

5.3. Disjoint-probe multilingual revision

The completed six-method, four-holdout, five-seed revision uses fixed test records and separate training probes and threshold-validation examples. Table 5 reports language-average and worst-language outcomes.

Relative to FedAvg, proxy-FLEN changes average micro-F_1 by -0.11 points (95% paired interval $[-0.64, 0.43]$) and worst-language micro-F_1 by $+0.30$ points (95% paired interval $[-0.13, 0.73]$). The frozen-backbone reference

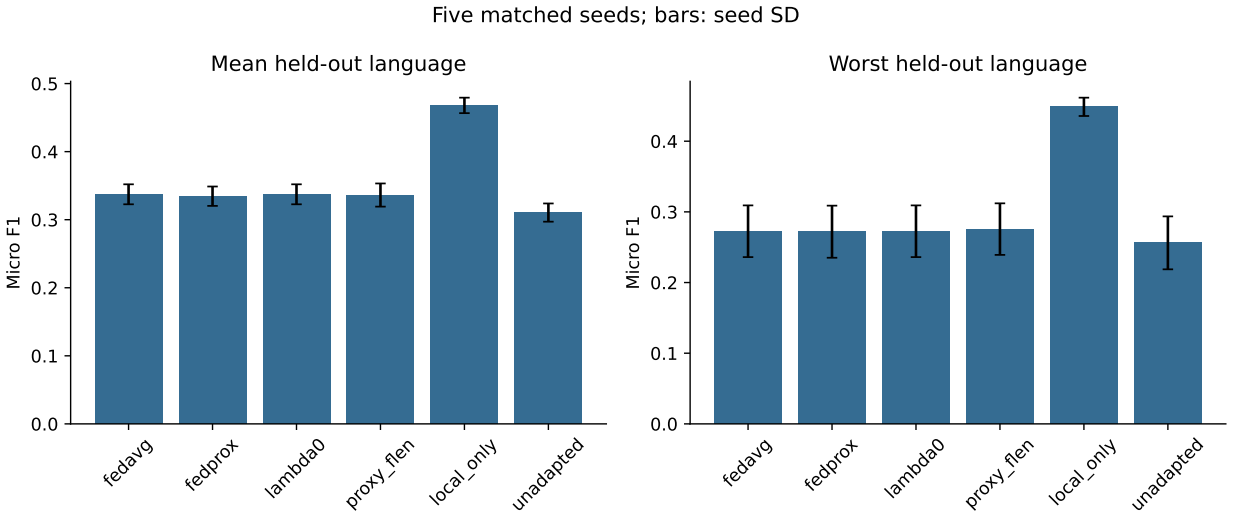


Figure 5: Mean and worst held-out-language performance in the completed revision; error bars show five-seed SD.

Table 6
Completed 15-run MultiEURLEX privacy revision over seeds 41–43. Held-out English–Polish results are means ± seed standard deviations. ECE pools binary label probabilities into 15 bins. The trusted-aggregator client-update accountant uses $\delta = 10^{-5}$; the attack measures record membership.

Condition	ϵ	Micro-F_1 ↑	Macro-F_1 ↑	BCE ↓	Brier ↓	ECE ↓	Attack AUC ↓
Unclipped	–	0.341 ± 0.039	0.184 ± 0.026	0.73 ± 0.04	0.222 ± 0.012	0.211 ± 0.050	0.617 ± 0.021
Clipping only	–	0.336 ± 0.039	0.181 ± 0.026	0.75 ± 0.04	0.227 ± 0.013	0.220 ± 0.051	0.615 ± 0.021
Gaussian 0.25	59.51	0.264 ± 0.058	0.196 ± 0.051	15.17 ± 5.71	0.488 ± 0.060	0.489 ± 0.060	0.591 ± 0.039
Gaussian 0.5	23.51	0.250 ± 0.046	0.206 ± 0.048	20.70 ± 5.65	0.481 ± 0.041	0.481 ± 0.041	0.525 ± 0.040
Gaussian 1.0	9.84	0.249 ± 0.051	0.210 ± 0.041	36.78 ± 10.68	0.479 ± 0.060	0.479 ± 0.059	0.543 ± 0.053

changes average micro-F_1 by -2.68 points (95% paired interval $[-3.96, -1.41]$); target-supervised local-only adaptation changes it by $+13.06$ points (95% paired interval $[11.91, 14.21]$). The zero-conflict control reproduces FedAvg. Proxy-FLEN's average contrast spans zero, so the study provides no stable incremental benefit from proxy conflict weighting. Target-language supervision remains substantially stronger than three-round source-language transfer. The frozen-backbone reference trains a classification head and therefore measures the contribution of backbone adaptation.

5.4. Client-update privacy and membership inference

The completed privacy revision compares unclipped training, clipping only, and Gaussian noise multipliers 0.25, 0.5, and 1.0 over seeds 41–43, giving 15 runs. It uses replace-one-client-update adjacency, an L_2 clipping norm of 1.0, full participation by three source-language clients, and a central Gaussian mechanism on the weighted aggregate. The mechanism assumes a trusted aggregator and protects the released update. The accountant reports the composed three-round (ϵ, δ) budget at $\delta = 10^{-5}$. Source-training probes are separated from threshold-validation records; final task evaluation uses held-out test records. A black-box loss attack compares source-language training members with source-language test non-members, separating attack calibration and evaluation halves. Its direction and threshold are selected on the calibration half. This diagnostic measures record membership, while the mechanism's protected unit is the client update.

Table 6 separates the clipping and noise effects. Clipping alone changes micro-F_1 from 0.341 to 0.336; the paired difference is -0.0056 with a 95% Student-t interval of $[-0.0173, 0.0061]$. At $\sigma = 0.5$, attack AUC is 0.525, compared with 0.615 for clipping only; its paired difference is -0.0901 ($[-0.2145, 0.0343]$). The corresponding micro-F_1 means are 0.250 and 0.336, while macro-F_1 rises from 0.181 to 0.206. The record-membership diagnostic therefore shows an uncertain attack reduction alongside differing task-metric responses.

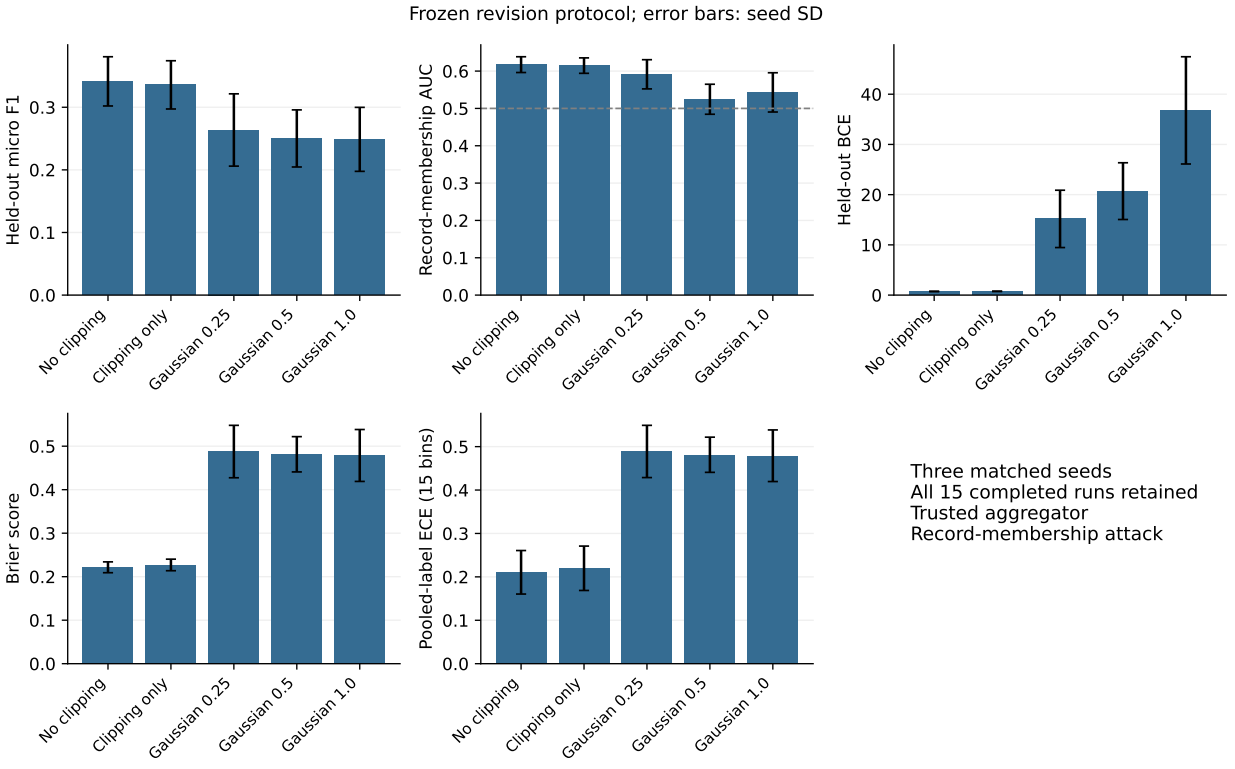


Figure 6: Completed privacy revision over three matched seeds. Bars show means and error bars show one seed standard deviation. The clipping-only control separates clipping from noise. Task performance, record-membership attack AUC, and probability quality are reported as distinct outcomes.

Probability quality deteriorates strongly under noise. At $\sigma = 0.5$, BCE rises from 0.75 to 20.70 and Brier score from 0.227 to 0.481 relative to clipping only; the paired Brier increase is 0.2544 ([0.1448, 0.3639]). Pooled binary-label ECE rises from 0.220 to 0.481, a paired increase of 0.2615 ([0.0563, 0.4666]). Figure 7 shows overconfident predictions across the noisy conditions. At $\sigma = 1.0$, attack AUC is 0.543 and BCE is 36.78, so the attack diagnostic does not improve monotonically with noise in this three-seed screen. All completed runs enter the analysis, including boundary-threshold runs. These findings characterize a poor probability-quality regime for the tested small-client, high-dimensional adapter mechanism.

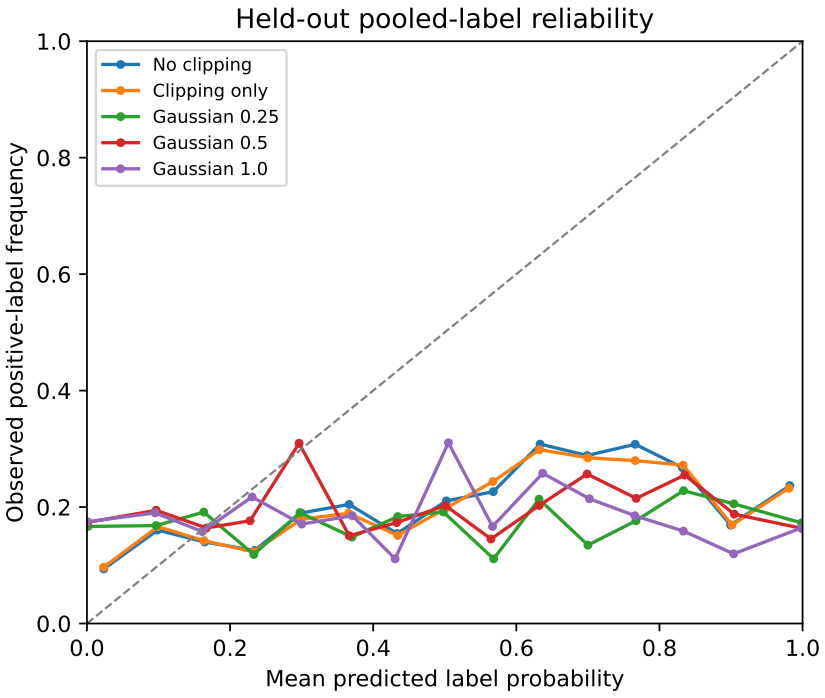


Figure 7: Held-out English–Polish reliability, pooling binary-label predictions across three seeds within each condition. Each point shows mean probability and observed positive frequency within a fixed probability bin. The dashed diagonal denotes perfect calibration; source data include bin counts.

5.5. Published-label legal component results

Eleven of twelve checkpoint conditions have completed the four-task diagnostic; the Mistral 7B reference remains pending model-weight recovery. Table 7 separates accuracy from class-balanced outcomes. The larger pretrained checkpoint differs in both model family and size; it provides a reference for component difficulty. The released seed and case contrasts include uncertainty, and probability metrics accompany each task.

Shared adaptation has task-dependent effects. FedAvg and FLEN reach 52.3% and 51.8% on overruling, compared with 49.2% for the base checkpoint. On citation support, their means fall to 49.3% and 49.1% from 53.7%. All three conditions produce the same UCC/common-law accuracy of 57.4% with balanced accuracy 50.0%, revealing a single-class solution. Personal-jurisdiction accuracy increases from 42.0% to 54.4% and 55.2%, while balanced accuracy remains below 56%. These outcomes identify limited component transfer and motivate expert authority annotation and independently verified citation retrieval.

Table 7
Published-label LegalBench component diagnostics on 492 fixed cases. Entries are percent; adapted rows average five shared checkpoints, pretrained rows evaluate one checkpoint.

Task	Checkpoint	Accuracy	Macro F1	Balanced accuracy
Overruling	Qwen base	49.2	43.8	51.5
Overruling	FedAvg shared	52.3	45.4	54.2
Overruling	FLEN shared	51.8	44.9	53.5
UCC/common law	Qwen base	57.4	36.5	50.0
UCC/common law	FedAvg shared	57.4	36.5	50.0
UCC/common law	FLEN shared	57.4	36.5	50.0
Citation support	Qwen base	53.7	47.9	53.7
Citation support	FedAvg shared	49.3	41.2	49.3
Citation support	FLEN shared	49.1	42.4	49.1
Personal jurisdiction	Qwen base	42.0	29.6	50.0
Personal jurisdiction	FedAvg shared	54.4	46.3	56.0
Personal jurisdiction	FLEN shared	55.2	46.4	55.2

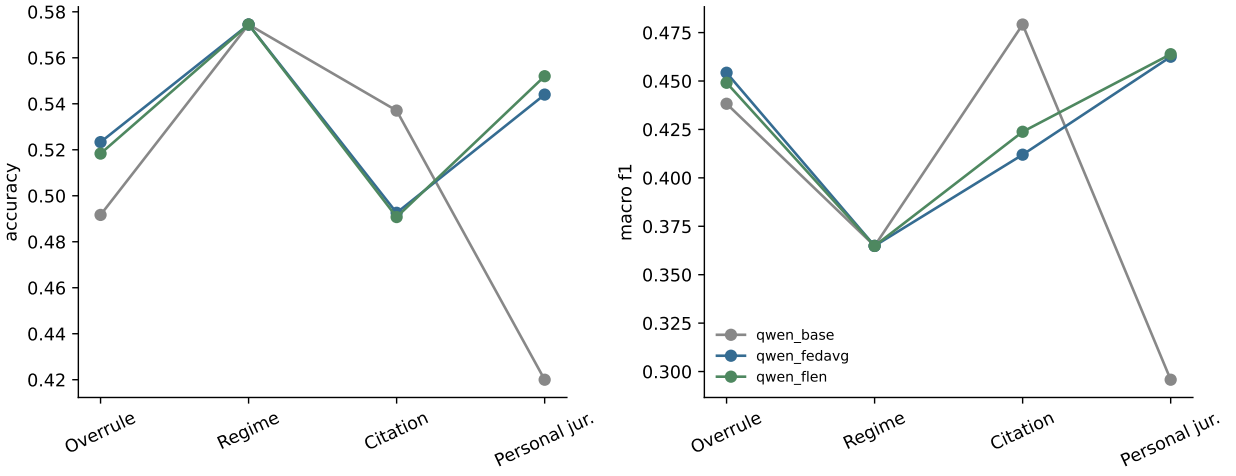


Figure 8: Fixed-case component diagnostics; adapted values average five training seeds. Tasks retain their published label spaces.

5.6. Connected shared-adapter reasoning results

The completed automatic comparison links exported FedAvg and FLEN shared adapters to the same inference protocol as the base checkpoint. Table 8 compares role prompting with one-call prediction and four-call self-refinement. These are paired case outcomes for seed-41 checkpoints. Legal soundness, coherence, preference and unsupported-claim rates await completed independent blinded review sheets.

The one-call FedAvg and FLEN controls parse successfully on only 14 and 10 of 60 cases, respectively. Their large four-stage gains therefore combine task accuracy and output-format compliance. Four-call self-refinement raises successful parsing to 58 and 59 cases. Against that control, four-stage accuracy changes by 0.00 points for FedAvg and 6.67 points for FLEN (95% paired bootstrap interval [0.00, 15.00]; exact McNemar $p = 0.219$). The base checkpoint decreases by 3.33 points. These call-matched results leave the reasoning advantage uncertain.

Table 8
Sixty paired reasoning cases per checkpoint. Differences are four-stage minus single-agent accuracy in points; intervals use paired case bootstrap. The four-stage comparator is regenerated deterministically across call-budget packages and counted once.

Checkpoint	Single calls	Single (%)	Four-stage (%)	Difference	95% CI	McNemar p
base	1	21.7	15.0	−6.7	[−16.7, 3.3]	0.344
base	4	18.3	15.0	−3.3	[−13.3, 6.7]	0.754
federated	1	5.0	20.0	+15.0	[5.0, 25.0]	0.012
federated	4	20.0	20.0	+0.0	[−5.0, 5.0]	1.000
flen	1	1.7	25.0	+23.3	[13.3, 35.0]	< 0.001
flen	4	18.3	25.0	+6.7	[0.0, 15.0]	0.219

Table 9
Output-format diagnostics on 60 cases. S denotes single-agent prediction and D four-stage deliberation. Conditional accuracy (percent) uses only cases parsed in both conditions and is descriptive; primary accuracy retains all cases and counts parsing failures as errors.

Checkpoint	Single calls	Parsed S	Parsed D	Both	Cond. acc. S	Cond. acc. D
base	1	58	55	53	18.9	17.0
base	4	58	55	54	16.7	16.7
federated	1	14	60	14	21.4	14.3
federated	4	58	60	58	20.7	19.0
flen	1	10	59	10	10.0	10.0
flen	4	59	59	58	19.0	24.1

Table 10
Mean analysis cost per case with four calls in both conditions. The common presentation step is excluded; call matching does not match token consumption.

Checkpoint	Protocol	Prompt tokens	Generated tokens	Seconds
base	Self-refinement	1942.7	96.7	1.68
base	Four-stage	2711.1	447.5	7.69
federated	Self-refinement	2074.6	251.0	5.70
federated	Four-stage	2635.5	375.6	8.51
flen	Self-refinement	2081.5	256.4	6.43
flen	Four-stage	2601.0	360.2	8.93

For the FLEN checkpoint, four-stage analysis consumes 360.3 generated tokens and 8.93 seconds per case, compared with 256.4 tokens and 6.43 seconds for self-refinement. The extra deliberation therefore increases generated tokens by about 41% and observed elapsed time by about 39%.

6. Discussion

What the matched experiments establish. Independent dual-LoRA experiments isolate a positive gate effect at both fixed training budgets: accuracy recovers by 2.50 and 2.64 points relative to unguarded personalization. The selected private branch adds little over the corresponding shared branch, indicating that screening harmful updates drives this

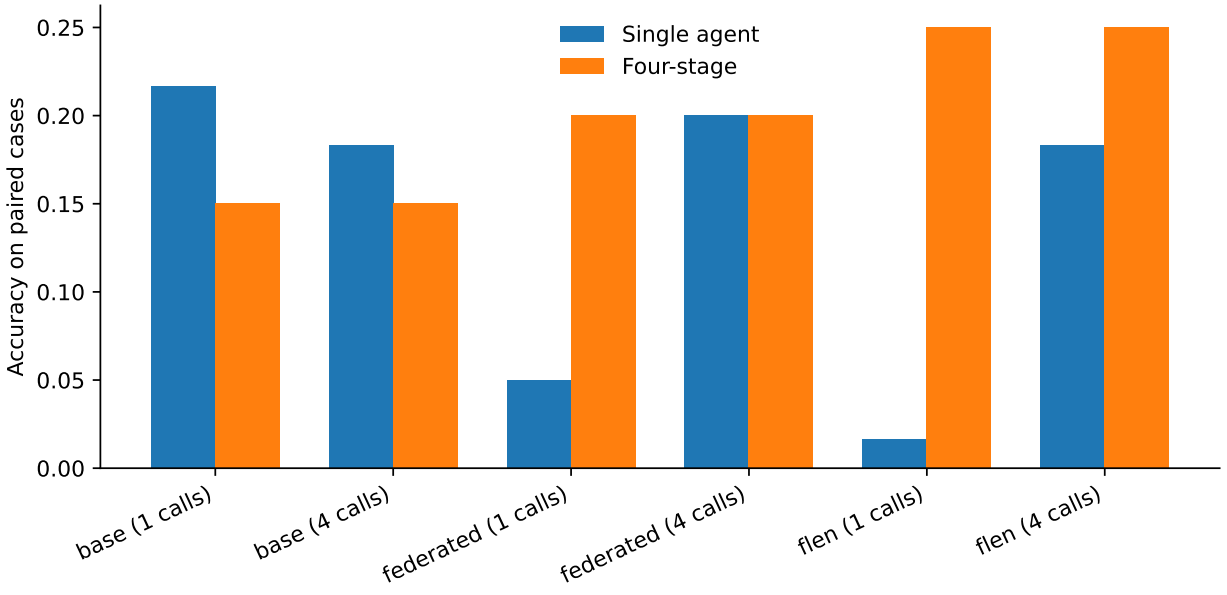


Figure 9: Automatic paired accuracy with base, FedAvg shared and FLEN shared checkpoints; call budgets label the single-agent controls.

recovery. Proxy conflict weighting adds only 0.06 points to guarded 8 + 2 training. FedAvg and FedProx remain competitive in average and fixed-shard worst-client accuracy. The five-seed intervals describe effect magnitudes, while exact sign-flip inference remains coarse at this replication level.

Authority scope and transfer. The central design principle is that legal heterogeneity includes sampling variation, task mismatch, model error, and legitimate authority-scoped disagreement. Comparability, target-conditioned cohorts, private adaptation, and a diversity floor provide a mechanism for treating these sources separately. The 120-run MultiEURLEX study shows an adaptation benefit over a frozen backbone and a substantial advantage for target-language supervision. Proxy-FLEN's mean micro-F_1 of 0.336 and FedAvg's 0.337 remain descriptively close, with an uncertain paired difference. CaseHOLD clients are heuristic cohorts and its test shards measure partition robustness; MultiEURLEX languages share EU authority. Expert-labelled authority pairs and naturally partitioned jurisdictions are therefore the next tests of the legal-conflict construct and minority-jurisdiction transfer.

Reasoning quality and resource use. Connecting shared federated adapters to inference exposes a format-compliance problem in the one-call controls. Four-call self-refinement largely resolves it and reduces the apparent deliberation advantage. At matched call counts, the FLEN checkpoint's four-stage gain is 6.67 points with exact McNemar $p = 0.219$, accompanied by roughly 41% more generated tokens. Published-label diagnostics also show uneven transfer: adapted checkpoints improve personal-jurisdiction accuracy but reduce citation-support accuracy. These findings motivate independent legal soundness and citation review, with two reviewers completing blinded sheets before adjudication. The current case-level analysis evaluates one trained checkpoint per method; replication across training seeds and a separate inference test set would strengthen generalization.

Privacy and probability quality. The privacy revision identifies a three-client regime with sharply increased probability loss, Brier score, and calibration error under noise. Record-membership AUC moves toward 0.5, with paired intervals spanning zero and a non-monotonic noise response. A formal budget for released client updates and an empirical record-membership score answer different questions. Future privacy comparisons should preserve that distinction, test update-targeted reconstruction under the stated observer, and select clipping and noise using development utility and calibration. Secure aggregation and inference-message protection require their own implementations and attack evaluations.

7. Ethics, Governance, and Intended Use

FLEN supports legal information processing and decision support under accountable human authority. Institutional use requires access to supporting authorities and uncertainty records, an appeal or override path, and documented responsibility for model and data governance. The system abstains when a rule lacks support or a jurisdictional conflict remains unresolved.

The measured pilot uses public benchmarks. Institutional studies document legal basis, data licences, retention, access control, cross-border restrictions, and the treatment of sealed or personally identifiable information. Reasoning summaries and audit trails may themselves be sensitive and require minimization, purpose limitation, and deletion policies.

Fairness evaluation includes minority jurisdictions and affected groups alongside aggregate utility. Qualified legal experts from the represented jurisdictions create or audit the authority mapping and conflict benchmark, with disagreement and uncertainty retained in the annotations. Privacy and policy auditors support traceability; legal oversight, independent security review, and empirical bias assessment remain required governance controls.

8. Conclusion

This paper presents FLEN, a data-local framework combining authority-scoped transfer, independent shared/private LoRA adaptation, and typed legal deliberation. Forty-five CaseHOLD runs establish matched-budget recovery from harmful personalization, with gate effects of 2.50 and 2.64 points across five seeds. The 120-run MultiEURLEX study shows useful source-language adaptation and a larger benefit from target-language supervision, while proxy conflict weighting has an uncertain incremental effect. Connected-adapter inference highlights format compliance and additional token cost as key factors in agent comparisons. The privacy study separates released-update protection, record-membership diagnostics, task utility, and probability calibration. Together, these results support guarded adaptation and provide a reproducible basis for expert authority validation, authentic-jurisdiction evaluation, and independent legal review.

Data and code availability

The manuscript source, implementation, experiment configurations, analysis scripts, and published result summaries are available at https://github.com/Robert-solo/federated_legal.git. The repository documents the software environment, random seeds, execution commands, and the correspondence between reported tables, figures, and source runs. The latest Elsevier CAS manuscript is the maintained paper version. Benchmark datasets are obtained from their original providers under their respective licences. Expert annotations and blinded-review outcomes are released with their provenance after human review is completed.

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